Command used:

For Docker:

1) sudo pacman -S docker docker-compose

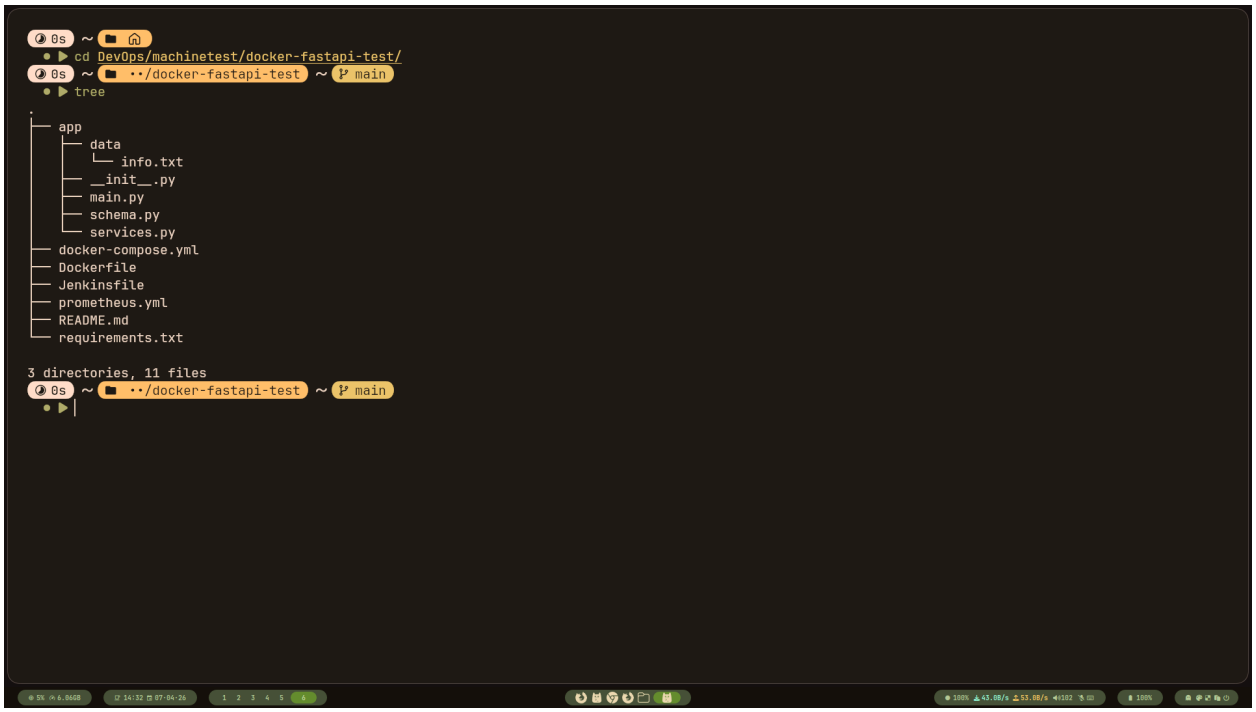
2) sudo systemctl start docker sudo systemctl enable docke

3) sudo usermod -aG docker $USER # (prathamesh)

4) docker --version

6) docker-compose --version

File and folder structure:



Command:
docker-compose up -d --build

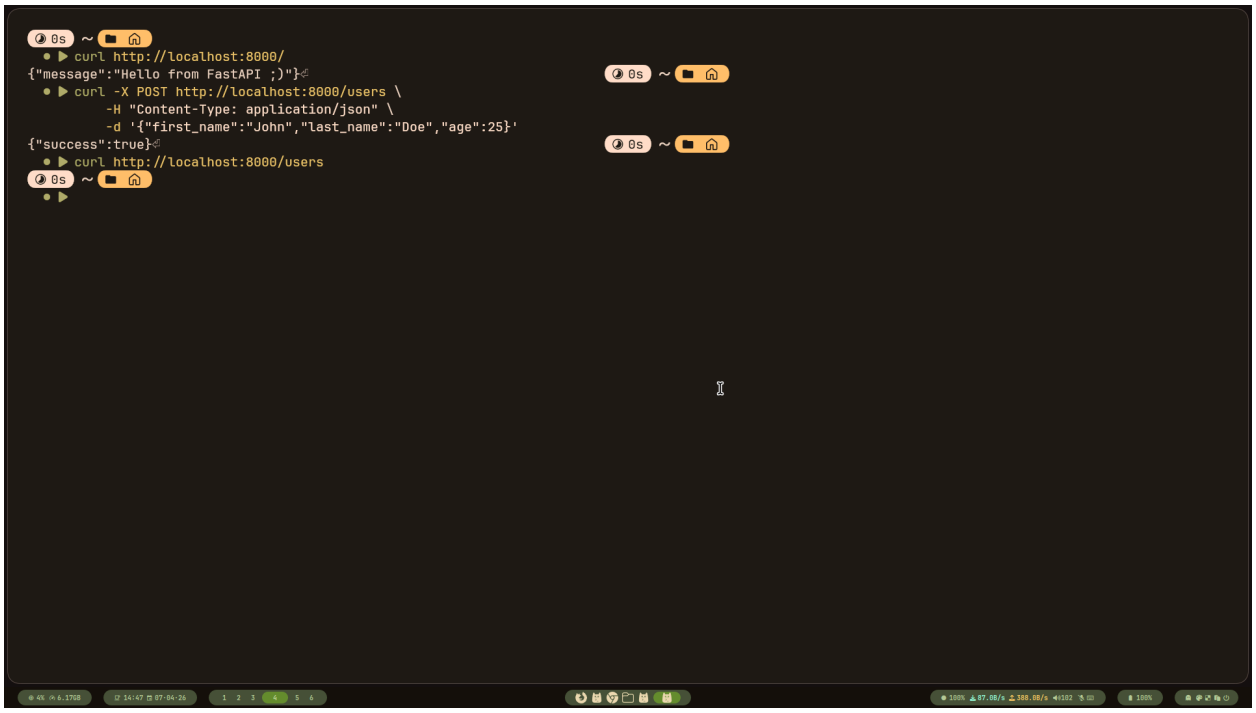
rebuild your custom images to include the latest code, start all defined services in the background, and free up terminal

```
ove it to avoid potential confusion
unable to get image 'prom/prometheus:latest': permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
0s ~ ••/docker-fastapi-test ~ main
  ✗ ▶ sudo usermod -aG docker $USER
[sudo] password for prathamesh:
4s ~ ••/docker-fastapi-test ~ main
  ● ▶ newgrp docker
```

```
❯ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
❯ docker-compose up -d --build
WARN[0000] /home/prathamesh/DevOps/machinetest/docker-fastapi-test/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please rem
ove it to avoid potential confusion
[+] up 18/24
[+] up 22/24fana/grafana:latest[⣿⣿⣿
```

1) **Test all 3 endpoints:**

```
curl http://localhost:8000/
{"message":"Hello from FastAPI ;)"}⏎

curl -X POST http://localhost:8000/users \
        -H "Content-Type: application/json" \
        -d '{"first_name":"John","last_name":"Doe","age":25}'
{"success":true}⏎

curl http://localhost:8000/users

{"data":[{"first_name":"John","last_name":"Doe","age":25},{"first_name":"John","last_name":"Doe
","age":25}]}⏎
```



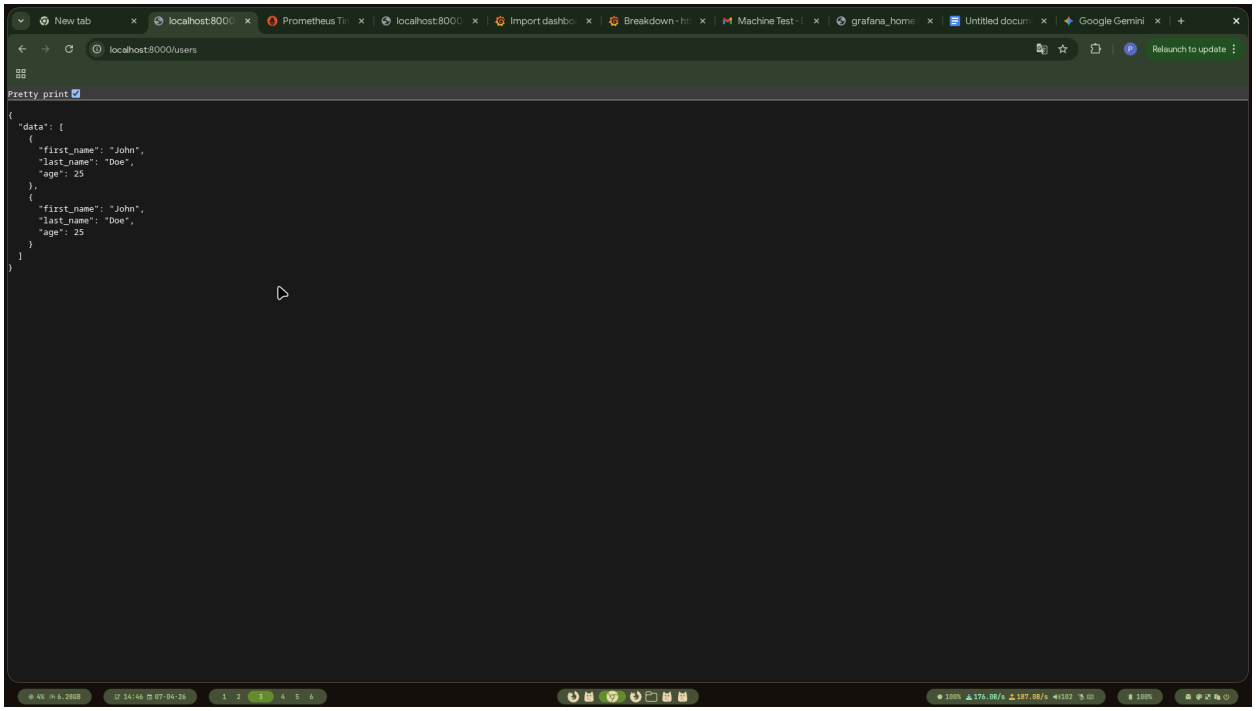
2) Test Data Persistence

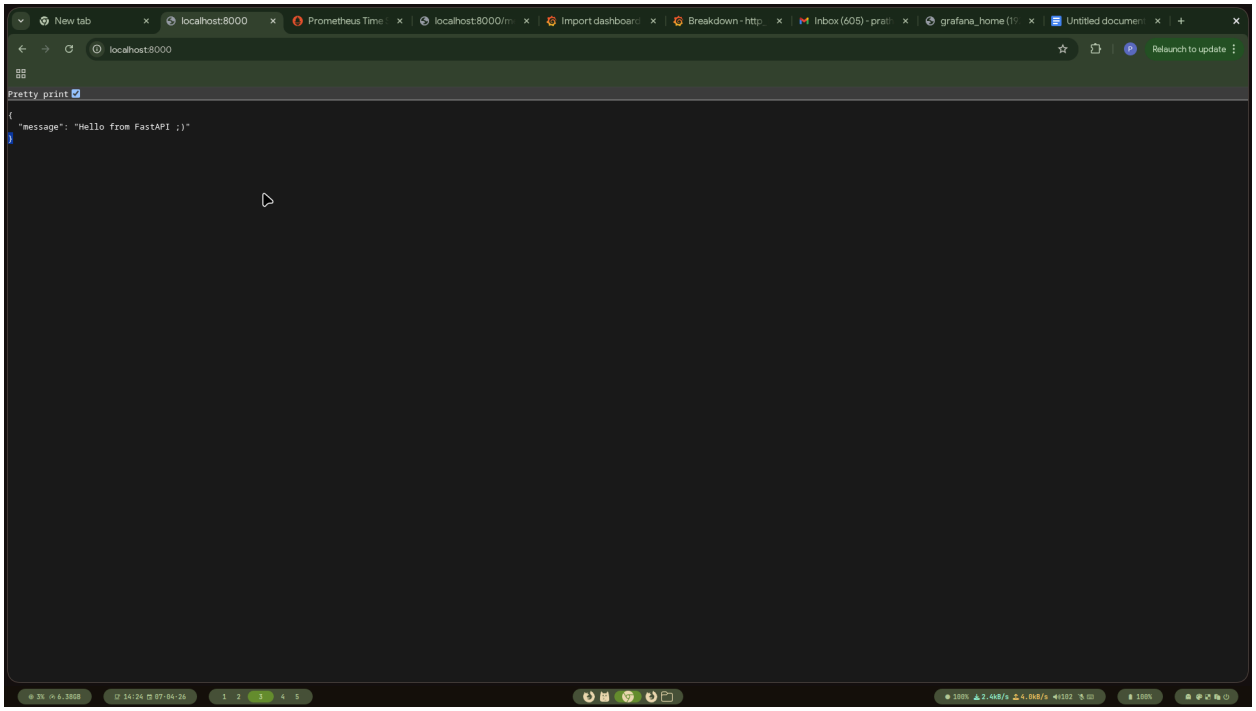
```
0s ~
curl -X POST http://localhost:8000/users \
        -H "Content-Type: application/json" \
        -d '{"first_name":"John","last_name":"Doe","age":25}'
{"success":true}
0s ~
curl http://localhost:8000/users
{"data":[{"first_name":"John","last_name":"Doe","age":25},{"first_name":"John","last_name":"Doe","age":25},{"first_name":"John","last_name":"Doe","age":25
}]}
0s ~
curl http://localhost:8000/users
{"data":[{"first_name":"John","last_name":"Doe","age":25},{"first_name":"John","last_name":"Doe","age":25},{"first_name":"John","last_name":"Doe","age":25
}]}
0s ~
docker-compose down
no configuration file provided: not found
0s ~
cd DevOps/machinetest/docker-fastapi-test/
0s ~ ••/docker-fastapi-test ~ main
docker-compose down
WARN[0000] /home/prathamesh/DevOps/machinetest/docker-fastapi-test/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please rem
ove it to avoid potential confusion
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
0s ~ ••/docker-fastapi-test ~ main
cd ~/DevOps/machinetest/docker-fastapi-test
0s ~ ••/docker-fastapi-test ~ main
docker-compose down
WARN[0000] /home/prathamesh/DevOps/machinetest/docker-fastapi-test/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please rem
ove it to avoid potential confusion
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
0s ~ ••/docker-fastapi-test ~ main
sudo usermod -aG docker $USER

[sudo] password for prathamesh:
Sorry, try again.
[sudo] password for prathamesh:
8s ~ ••/docker-fastapi-test ~ main
newgrp docker
```

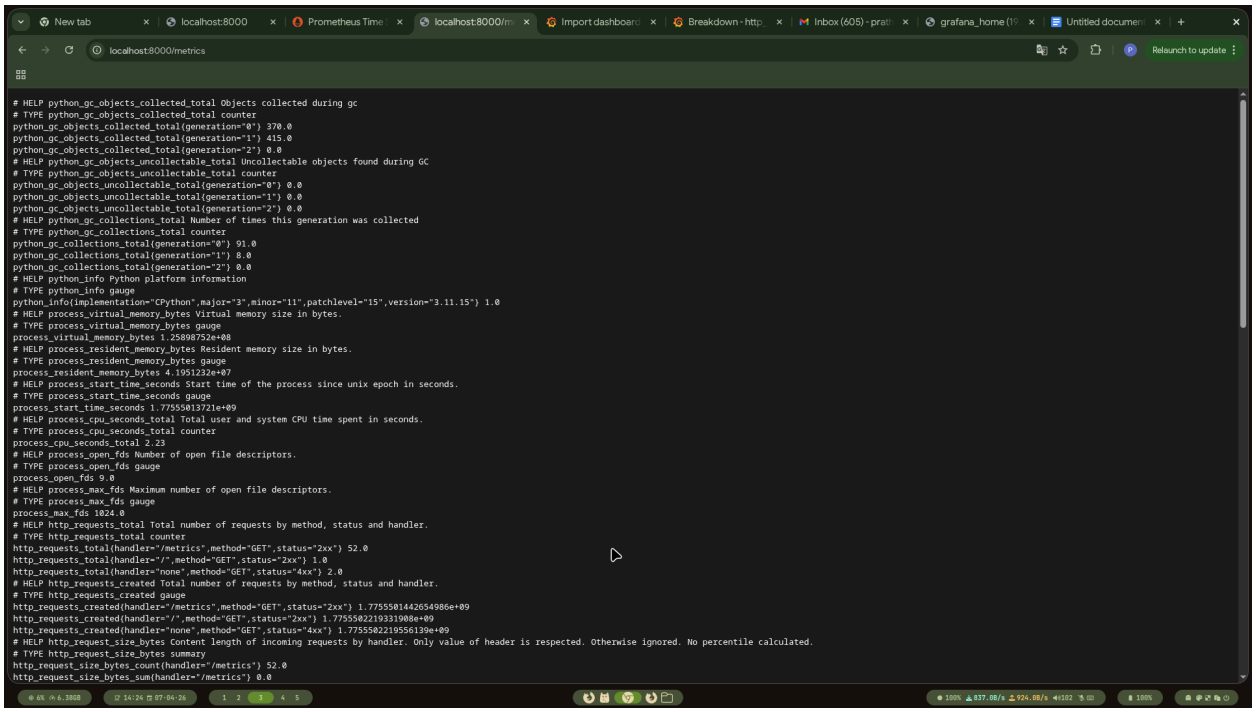
```
❯ docker-compose down
WARN[0000] /home/prathamesh/DevOps/machinetest/docker-fastapi-test/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please rem
ove it to avoid potential confusion
[+] down 4/4
 ✔ Container grafana                  Removed                                                                                    0.4s
 ✔ Container fastapi_app              Removed                                                                                    0.5s
 ✔ Container prometheus               Removed                                                                                    0.2s
 ✔ Network docker-fastapi-test_default Removed                                                                                   0.2s
❯ docker-compose up -d
WARN[0000] /home/prathamesh/DevOps/machinetest/docker-fastapi-test/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please rem
ove it to avoid potential confusion
[+] up 4/4
 ✔ Network docker-fastapi-test_default Created                                                                                   0.1s
 ✔ Container prometheus               Started                                                                                    0.4s
 ✔ Container fastapi_app              Started                                                                                    0.4s
 ✔ Container grafana                  Started                                                                                    0.5s
```

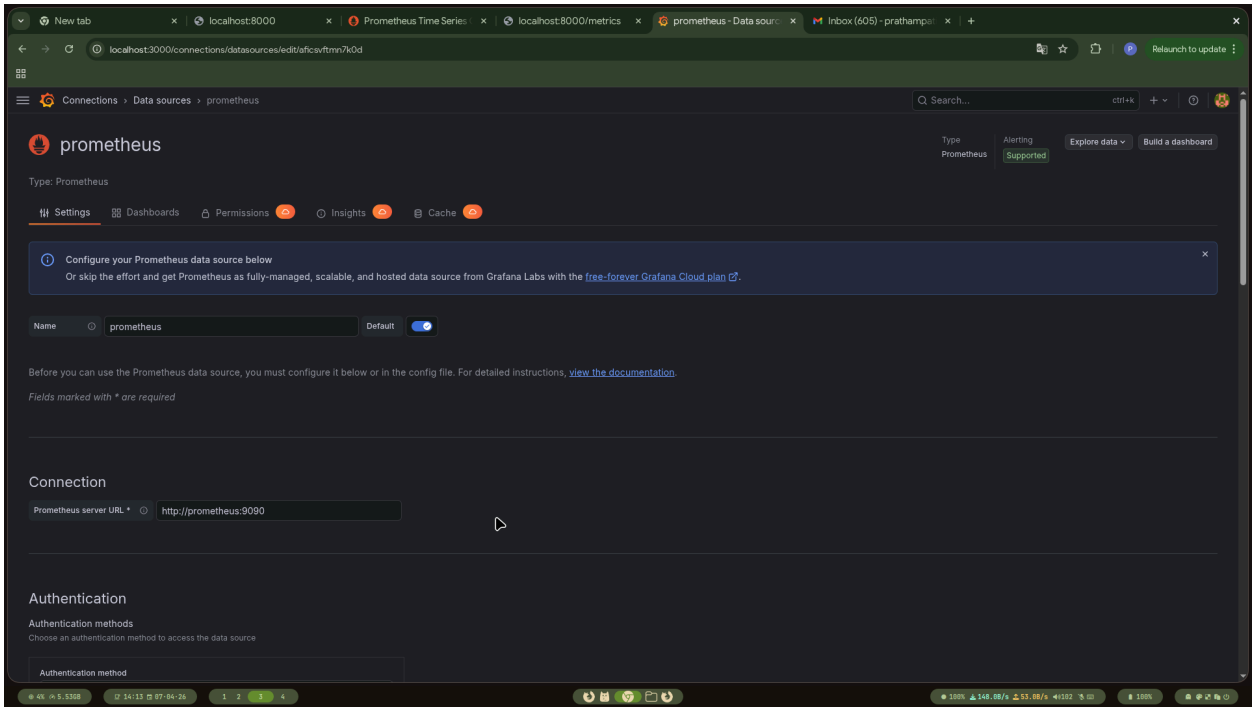
```
 ✔ Network docker-fastapi-test_default Removed                                                                                   0.2s
❯ docker-compose up -d
WARN[0000] /home/prathamesh/DevOps/machinetest/docker-fastapi-test/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please rem
ove it to avoid potential confusion
[+] up 4/4
 ✔ Network docker-fastapi-test_default Created                                                                                   0.1s
 ✔ Container prometheus               Started                                                                                    0.4s
 ✔ Container fastapi_app              Started                                                                                    0.4s
 ✔ Container grafana                  Started                                                                                    0.5s
❯ curl http://localhost:8000/users\
>
❯ curl http://localhost:8000/users
{"data":[{"first_name":"John","last_name":"Doe","age":25},{"first_name":"John","last_name":"Doe","age":25},{"first_name":"John","last_name":"Doe","age":25
}]}%

~/DevOps/machinetest/docker-fastapi-test  main !2 ?4
```

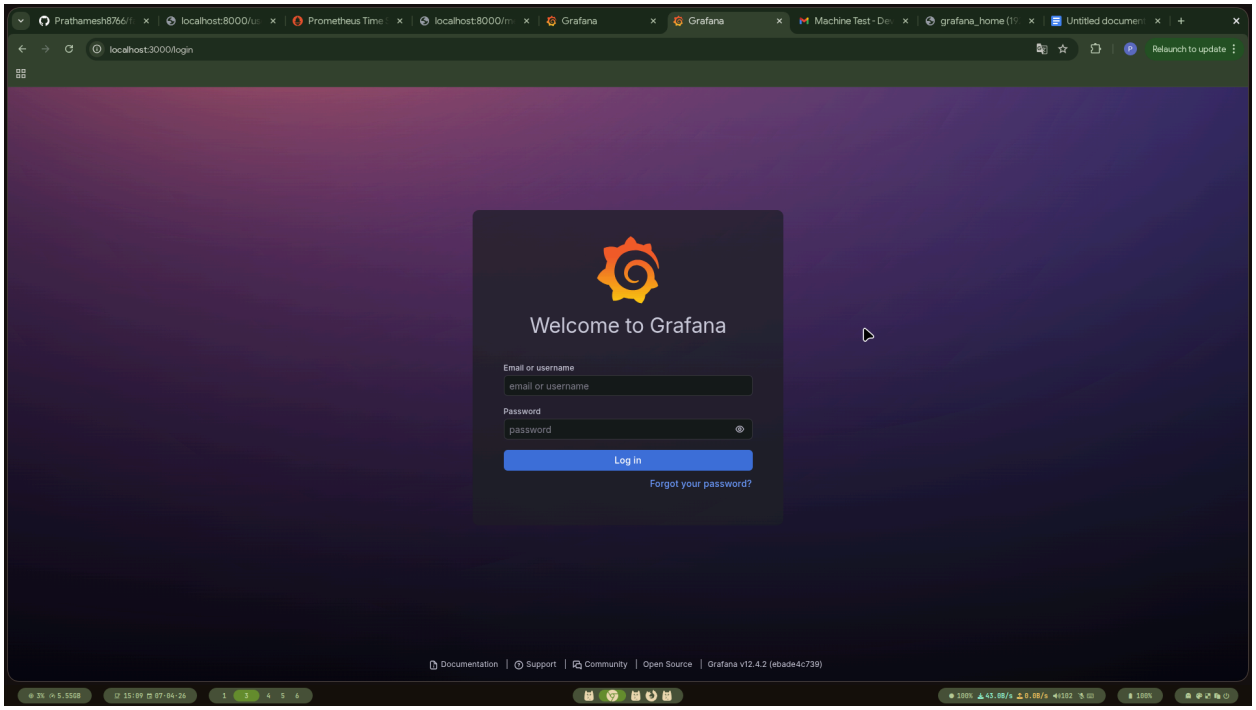
OutPuts:

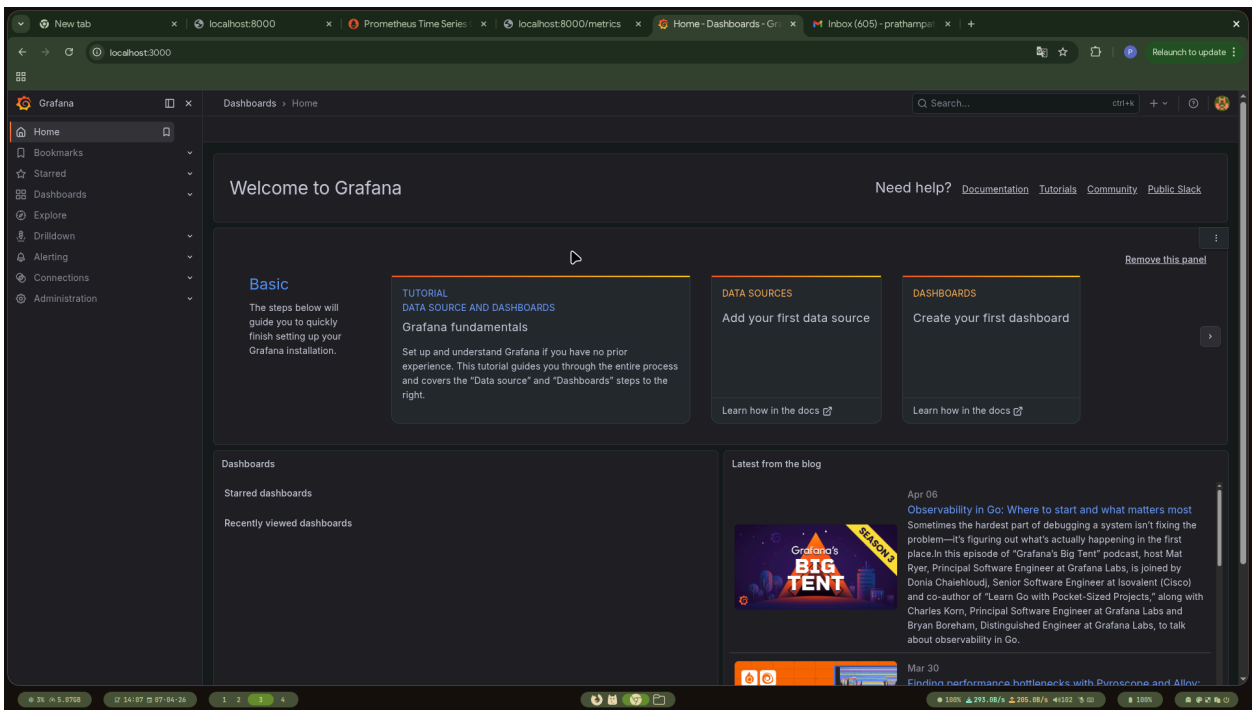

localhost:8000/users
Pretty print
{
"data": [
{
"first_name": "John",
"last_name": "Doe",
"age": 25
},
{
"first_name": "John",
"last_name": "Doe",
"age": 25
}
]
}

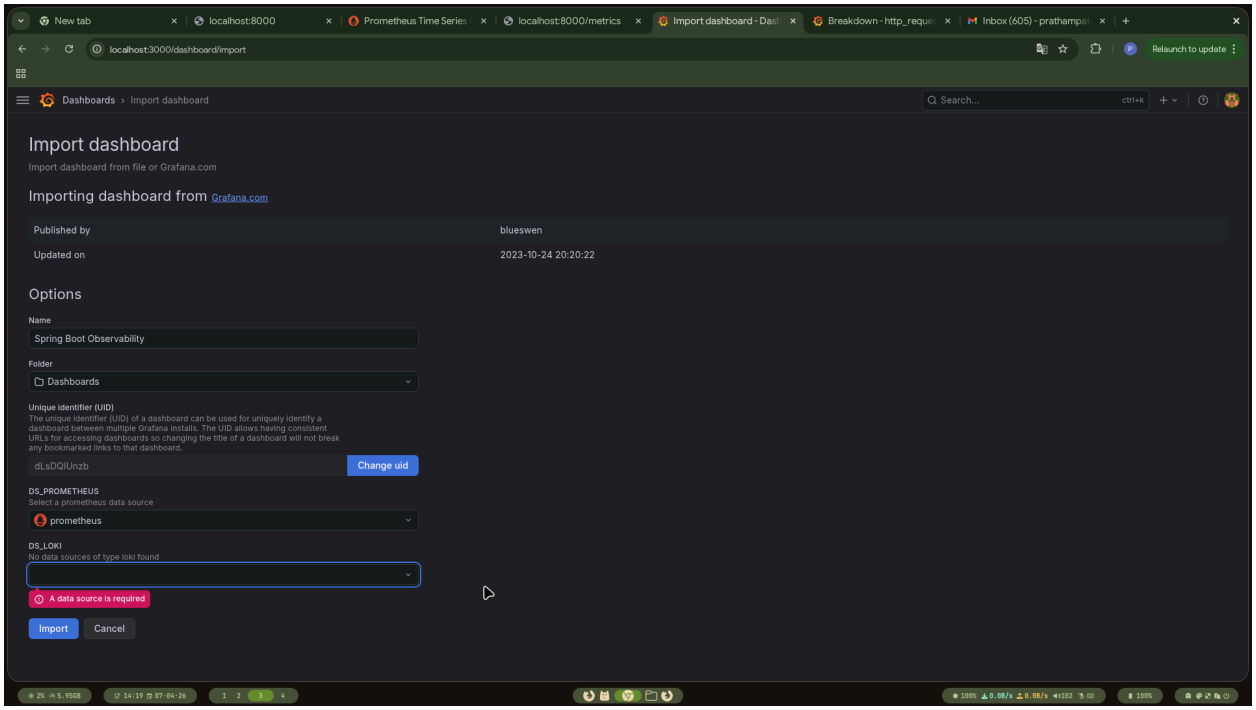

localhost:8000
Pretty print
{
"message": "Hello from FastAPI ;)"
}

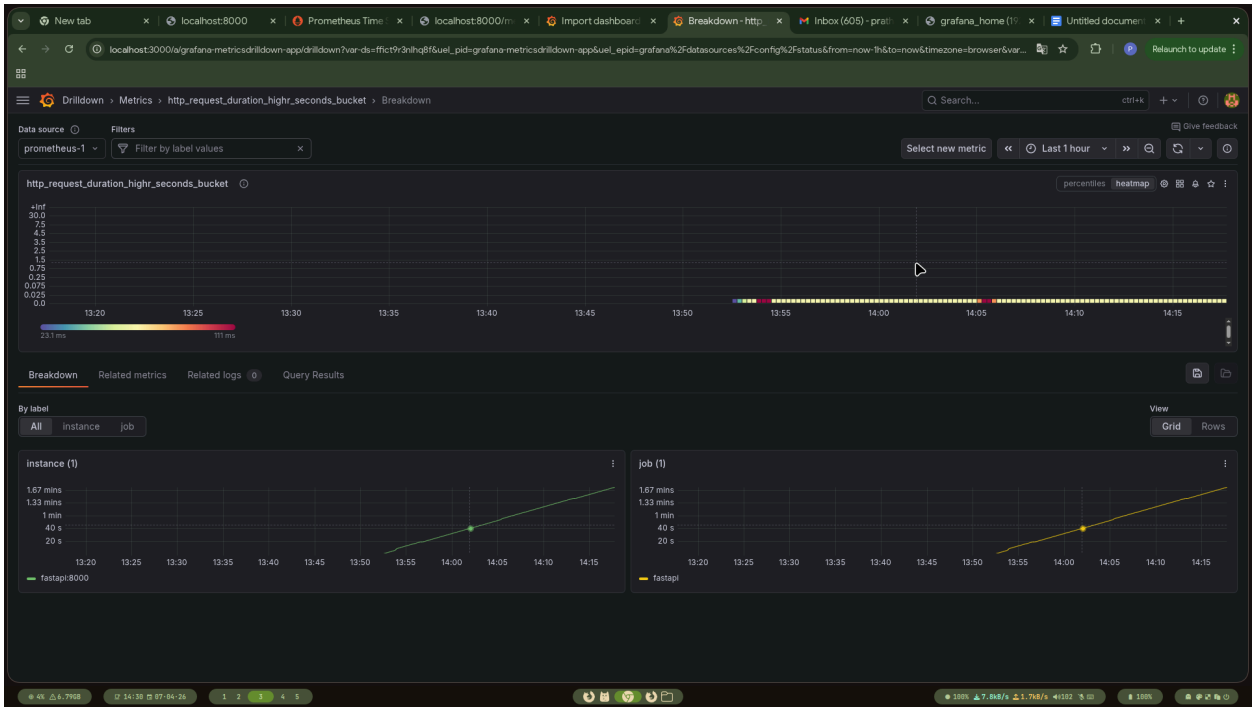

New tab
localhost:8000
Prometheus Time
localhost:8000/m
Import dashboard
Breakdown - http_
Inbox (605) - prath
grafana_home (19
Untitled documen
localhost:8000/metrics
Relaunch to update
HELP python_gc_objects_collected_total Objects collected during gc
TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 370.0
python_gc_objects_collected_total{generation="1"} 415.0
python_gc_objects_collected_total{generation="2"} 0.0
HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
HELP python_gc_collections_total Number of times this generation was collected
TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 91.0
python_gc_collections_total{generation="1"} 8.0
python_gc_collections_total{generation="2"} 0.0
HELP python_info Python platform information
TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="11",patchlevel="15",version="3.11.15"} 1.0
HELP process_virtual_memory_bytes Virtual memory size in bytes.
TYPE process_virtual_memory_bytes gauge
process_virtual_memory_bytes 1.25898752e+08
HELP process_resident_memory_bytes Resident memory size in bytes.
TYPE process_resident_memory_bytes gauge
process_resident_memory_bytes 4.1951232e+07
HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
TYPE process_start_time_seconds gauge
process_start_time_seconds 1.77555013721e+09
HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
TYPE process_cpu_seconds_total counter
process_cpu_seconds_total 2.23
HELP process_open_fds Number of open file descriptors.
TYPE process_open_fds gauge
process_open_fds 9.0
HELP process_max_fds Maximum number of open file descriptors.
TYPE process_max_fds gauge
process_max_fds 1024.0
HELP http_requests_total Total number of requests by method, status and handler.
TYPE http_requests_total counter
http_requests_total{handler="/metrics",method="GET",status="2xx"} 52.0
http_requests_total{handler="/",method="GET",status="2xx"} 1.0
http_requests_total{handler="none",method="GET",status="4xx"} 2.0
HELP http_requests_created Total number of requests by method, status and handler.
TYPE http_requests_created gauge
http_requests_created{handler="/metrics",method="GET",status="2xx"} 1.7755501442654986e+09
http_requests_created{handler="/",method="GET",status="2xx"} 1.7755502219331908e+09
http_requests_created{handler="none",method="GET",status="4xx"} 1.7755502219556139e+09
HELP http_request_size_bytes Content length of incoming requests by handler. Only value of header is respected. Otherwise ignored. No percentile calculated.
TYPE http_request_size_bytes summary
http_request_size_bytes_count{handler="/metrics"} 52.0
http_request_size_bytes_sum{handler="/metrics"} 0.0


New tab
localhost:8000
Prometheus Time Series
localhost:8000/metrics
prometheus - Data sourc
Inbox (605) - prathampa
localhost:3000/connections/datasources/edit/aficsvftmn7k0d
Relaunch to update
Connections › Data sources › prometheus
Search...
ctrl+k
prometheus
Type
Prometheus
Alerting
Supported
Explore data
Build a dashboard
Type: Prometheus
Settings
Dashboards
Permissions
Insights
Cache
Configure your Prometheus data source below
Or skip the effort and get Prometheus as fully-managed, scalable, and hosted data source from Grafana Labs with the free-forever Grafana Cloud plan.
Name
prometheus
Default
Before you can use the Prometheus data source, you must configure it below or in the config file. For detailed instructions, view the documentation.
Fields marked with * are required
Connection
Prometheus server URL *
http://prometheus:9090
Authentication
Authentication methods
Choose an authentication method to access the data source
Authentication method


localhost:3000/login
Welcome to Grafana
Email or username
email or username
Password
password
Log in
Forgot your password?
Documentation | Support | Community | Open Source | Grafana v12.4.2 (ebade4c739)

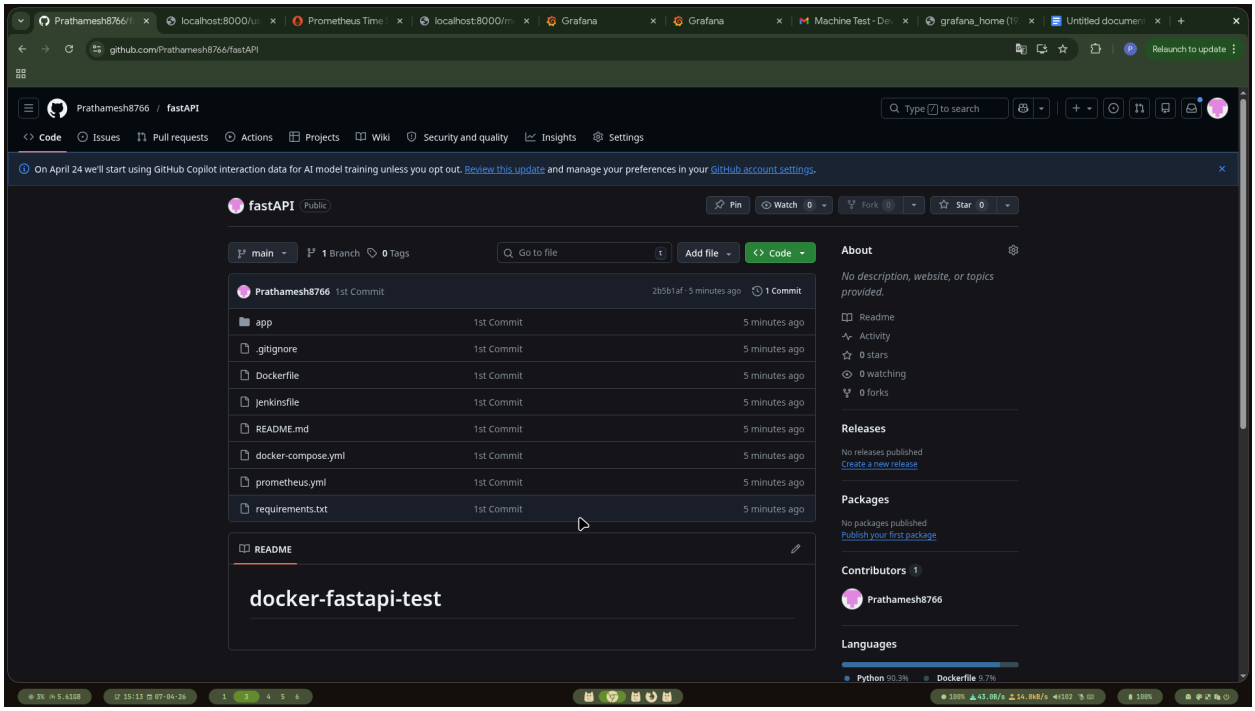

localhost:3000
Grafana
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Administration
Dashboards › Home
Welcome to Grafana
Need help? Documentation Tutorials Community Public Slack
Remove this panel
Basic
The steps below will guide you to quickly finish setting up your Grafana installation.
TUTORIAL
DATA SOURCE AND DASHBOARDS
Grafana fundamentals
Set up and understand Grafana if you have no prior experience. This tutorial guides you through the entire process and covers the "Data source" and "Dashboards" steps to the right.
DATA SOURCES
Add your first data source
Learn how in the docs
DASHBOARDS
Create your first dashboard
Learn how in the docs
Dashboards
Starred dashboards
Recently viewed dashboards
Latest from the blog
Apr 06
Observability in Go: Where to start and what matters most
Sometimes the hardest part of debugging a system isn't fixing the problem—it's figuring out what's actually happening in the first place.In this episode of "Grafana's Big Tent" podcast, host Mat Ryer, Principal Software Engineer at Grafana Labs, is joined by Donia Chaiehloudj, Senior Software Engineer at Isovalent (Cisco) and co-author of "Learn Go with Pocket-Sized Projects," along with Charles Korn, Principal Software Engineer at Grafana Labs and Bryan Boreham, Distinguished Engineer at Grafana Labs, to talk about observability in Go.
Grafana's BIG TENT
SEASON 3
Mar 30


New tab
localhost:8000
Prometheus Time Series
localhost:8000/metrics
Import dashboard - Dash
Breakdown - http_reque
Inbox (605) - prathampa
localhost:3000/dashboard/import
Relaunch to update
Dashboards › Import dashboard
Search...
Import dashboard
Import dashboard from file or Grafana.com
Importing dashboard from Grafana.com
Published by
blueswen
Updated on
2023-10-24 20:20:22
Options
Name
Spring Boot Observability
Folder
Dashboards
Unique identifier (UID)
The unique identifier (UID) of a dashboard can be used for uniquely identify a dashboard between multiple Grafana installs. The UID allows having consistent URLs for accessing dashboards so changing the title of a dashboard will not break any bookmarked links to that dashboard.
dLsDQIUnzb
Change uid
DS_PROMETHEUS
Select a prometheus data source
prometheus
DS_LOKI
No data sources of type loki found
A data source is required
Import
Cancel


New tab
localhost:8000
Prometheus Time
localhost:8000/m
Import dashboard
Breakdown - http
Inbox (605) - prat
grafana_home (19
Untitled documen
localhost:3000/a/grafana-metricsdrilldown-app/drilldown?var-ds=ffict9r3nlhq8f&uel_pid=grafana-metricsdrilldown-app&uel_epid=grafana%2Fdatasources%2Fconfig%2Fstatus&from=now-1h&to=now&timezone=browser&var...
Relaunch to update
Drilldown › Metrics › http_request_duration_highr_seconds_bucket › Breakdown
Search...
Data source
prometheus-1
Filters
Filter by label values
Give feedback
Select new metric
Last 1 hour
http_request_duration_highr_seconds_bucket
percentiles
heatmap
Breakdown
Related metrics
Related logs
Query Results
By label
All
instance
job
View
Grid
Rows
instance (1)
job (1)
fastapi:8000
fastapi

Git :

```
0s ~ ··/DevOps/machinetest ~ main
  ● ▶ cd docker-fastapi-test/
0s ~ ··/docker-fastapi-test ~ main
  ● ▶ ls -a
.git        app                 Dockerfile   ! prometheus.yml   requirements.txt
.gitignore  docker-compose.yml  Jenkinsfile  README.md
0s ~ ··/docker-fastapi-test ~ main
  ● ▶ rm -rf .git
0s ~ ··/DevOps/machinetest/docker-fastapi-test ~ main
  ● ▶ ls -
"-": No such file or directory (os error 2)
0s ~ ··/DevOps/machinetest/docker-fastapi-test ~ main
  ✕ ▶ ls -a
.gitignore  docker-compose.yml  Jenkinsfile     README.md
app         Dockerfile          ! prometheus.yml  requirements.txt
0s ~ ··/DevOps/machinetest/docker-fastapi-test ~ main
  ● ▶
```

```
  ● ▶ git commit -m "1st Commit"
On branch master

Initial commit

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        Dockerfile
        Jenkinsfile
        README.md
        app/
        docker-compose.yml
        prometheus.yml
        requirements.txt

nothing added to commit but untracked files present (use "git add" to track)
0s ~ ··/docker-fastapi-test ~ master
  ✕ ▶ git add .
0s ~ ··/docker-fastapi-test ~ master
  ● ▶ git commit -m "1st Commit"
[master (root-commit) 2b5b1af] 1st Commit
 12 files changed, 310 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 Dockerfile
 create mode 100644 Jenkinsfile
 create mode 100644 README.md
 create mode 100644 app/__init__.py
 create mode 100644 app/data/info.txt
 create mode 100644 app/main.py
 create mode 100644 app/schema.py
 create mode 100644 app/services.py
 create mode 100644 docker-compose.yml
 create mode 100644 prometheus.yml
 create mode 100644 requirements.txt
0s ~ ··/docker-fastapi-test ~ master
```

```
 create mode 100644 Dockerfile
 create mode 100644 Jenkinsfile
 create mode 100644 README.md
 create mode 100644 app/__init__.py
 create mode 100644 app/data/info.txt
 create mode 100644 app/main.py
 create mode 100644 app/schema.py
 create mode 100644 app/services.py
 create mode 100644 docker-compose.yml
 create mode 100644 prometheus.yml
 create mode 100644 requirements.txt
0s ~ ••/docker-fastapi-test ~ master
  git branch -M main
0s ~ ••/docker-fastapi-test ~ main
  git remote add origin git@github.com:Prathamesh8766/fastAPI.git
0s ~ ••/docker-fastapi-test ~ main
  git push -u origin mai
error: src refspec mai does not match any
error: failed to push some refs to 'github.com:Prathamesh8766/fastAPI.git'
0s ~ ••/docker-fastapi-test ~ main
  git push -u origin mai
error: src refspec mai does not match any
error: failed to push some refs to 'github.com:Prathamesh8766/fastAPI.git'
0s ~ ••/docker-fastapi-test ~ main
  git push -u origin main
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 16 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (16/16), 3.57 KiB | 1.78 MiB/s, done.
Total 16 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Prathamesh8766/fastAPI.git
 * [new branch]      main → main
branch 'main' set up to track 'origin/main'.
4s ~ ••/docker-fastapi-test ~ main
```